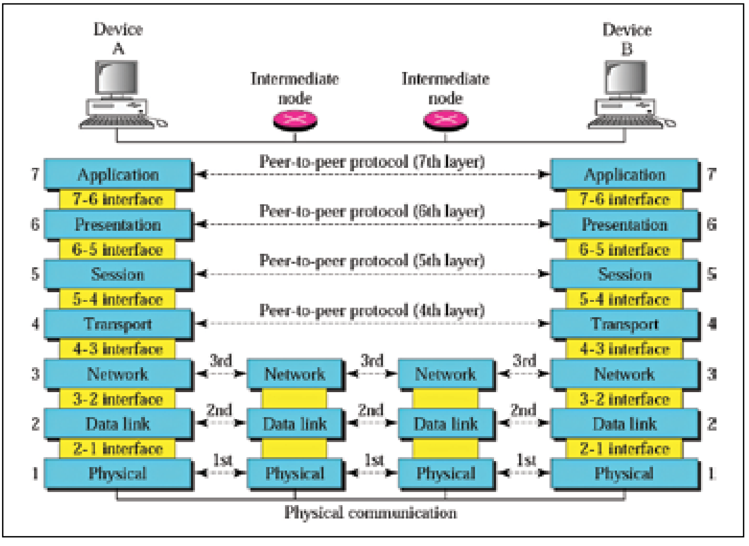
Consider this scenario, you are browsing a website using *HTTPS* on your tablet while it is connected to your phone's Bluetooth *PAN* and the phone is connected to the internet via *LTE*. In this case, the tablet is sending and receiving *HTTP* data using *TCP* protocol where the data was encrypted on the session layer of the *OSI* model and was wrapped and routed using the Bluetooth protocol by the tablet. This data was then unwrapped on the phone and re-sent using the *LTE* protocol. If the protocol stacks for *HTTP*, *TCP*, *Bluetooth* or *LTE* were not implemented, it wouldn't have been possible to achieve this feat!

Software creation for SoC

To write software that works with an SoC, one needs the right set of development tools, just the way you need Android Studio (in fact, the toolchain it provides) for building Android apps. For writing an SoC software, you would need a compiler which can convert your code to binary format understood by the processor. Most lower level SoC software is written in C or C++ and is compiled for the specific platform. You would need a C compiler for ARM architecture if you need to build a program that runs on Raspberry Pi but you would need the compiler for x86 architecture if you are building for Intel Galileo platform. Drivers or protocol stacks for an SoC are also written in C using the development toolchain for that architecture.



The OSI Protocol stack